Performance Analysis of Probabilistically Shaped CV-QKD in Free-space HAP-to-UAV Systems

Binh Nguyen*, Cuong T. Nguyen†, Hoang D. Le†, and Trinh Van Chien*
*School of Information and Communications Technology,
Hanoi University of Science and Technology, Hanoi 100000, Vietnam
†Computer Communications Laboratory, The University of Aizu, Aizuwakamatsu 965-8580, Japan

***Abstract*—Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.**

***Index Terms*—Continuous-variable quantum key distribution (CV–QKD), unmanned aerial vehicle (UAV), free-space optics (FSO), high-altitude platform (HAP), pulse constellation shaping (PCS)**

I. Introduction

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

II. System Model

A. System Description & Signal Model

Fig. 1 depicts the considered FSO/QKD system between a HAP (Alice) and a UAV (Bob) enabling secure IoV applications. Therein, we adopt the DM-CV-QKD protocol with M-

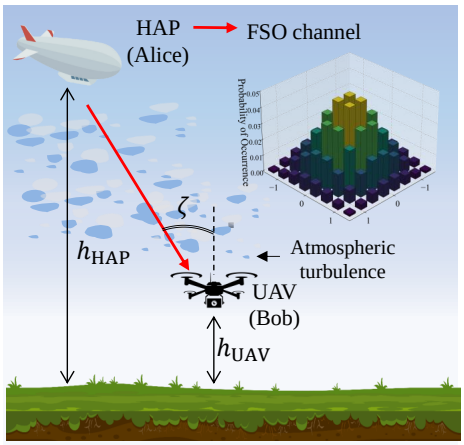


Fig. 1. The considered HAP-to-UAV FSO/CV-QKD systems enabling secure IoV applications.

QAM modulation. In this modulation scheme, Alice randomly selects a coherent state from a deterministic set $\{|\alpha_{\mathbf{k}}\rangle\}$, where $\mathbf{k} \in \mathcal{K} = \{(i,j) : i,j \in \{0,\ldots,m-1\}\}$, $m = \sqrt{M}$. The value $\alpha_{\mathbf{k}}$ is given as

$$\alpha_{\mathbf{k}} = \Delta\left(i - \frac{m-1}{2}\right) + j_{\text{c}}\Delta\left(j - \frac{m-1}{2}\right), \tag{1}$$

where Δ is the symbol spacing, and $j_{\text{c}} = \sqrt{-1}$ denotes the imaginary unit. The coherent state $|\alpha_{\mathbf{k}}\rangle$ is selected with probability $p_{\mathbf{k}}$, which follows the Maxwell–Boltzmann (MB) distribution, and is given as [1]

$$p_{\mathbf{k}} = \frac{\exp\left(-\gamma|\mathbf{k}|^2\right)}{\mathcal{Z}}, \tag{2}$$

where $\mathcal{Z} = \sum_{\mathbf{k}\in\mathcal{K}} \exp\left[-\nu|\mathbf{k}|^2\right]$ is the normalization factor, and γ is the shaping parameter. It is observed that setting $\gamma = 0$ results in uniform probability among coherent states. Finally, the average photon number of the system, $\langle n\rangle$, can be calculated as [2]

$$\langle n\rangle = \sum_{\mathbf{k}\in\mathcal{K}} p_{\mathbf{k}}|\alpha_{\mathbf{k}}|^2, \tag{3}$$

where $V_{\text{mod}} = \langle n\rangle/2$ refer to the modulation variance.

B. Quantum Channel Model

The transmittance of the free-space HAP-to-UAV channel can be expressed as $\eta = \eta_{\text{atm}} I_{\text{a}} \eta_{\text{p}} \eta_{\text{AoA}}$, where η_{atm} is the deterministic atmospheric loss, I_{a} is the atmospheric-induced turbulence, η_{p} denotes the geometric loss and misequationment, and η_{AoA} denotes the angle-of-arrival (AoA) fluctuation. *Regarding the atmospheric loss,* it can be given as

$$\eta_{\text{atm}} = \tau_{\text{zen}}^{\sec(\zeta)}, \tag{4}$$

where ζ is the zenith angle, and $\tau_{\text{zen}} \in [0,1]$ is the atmospheric transmittance at zenith.

For the atmospheric turbulence, by assuming the weak turbulence condition, we can model I_{a} as a log-normal random variable, whose probability density function (PDF) is [3]

$$f_{I_{\text{a}}}(I_{\text{a}}) = \frac{1}{I_{\text{a}}\sqrt{2\pi\sigma_{\text{R}}^2}} \exp\left[-\frac{\left(\ln I_{\text{a}} + \frac{\sigma_{\text{R}}^2}{2}\right)^2}{2\sigma_{\text{R}}^2}\right], \tag{5}$$

where $\sigma_{\text{R}}^2 = 2.25\left(\frac{2\pi}{\lambda}\right)^{\frac{7}{6}} \sec^{\frac{11}{6}}(\zeta) \int_{h_{\text{UAV}}}^{h_{\text{atm}}} C_{\text{n}}^2(h)\,(h - h_{\text{UAV}})^{\frac{5}{6}}\,\text{d}h$ is the Rytov variance, h_{atm} is the atmospheric altitude, h_{UAV} denotes the UAV's altitude, and λ is the optical wavelength. In addition, $C_{\text{n}}^2(h)$ can be computed as a function of the ground turbulence level, $C_{\text{n}}^2(0)$, and the root-mean-square (rms) wind speed, v_{wind}, as in [].

As for geometric loss and misalignment, the fraction of collected power given the distance between the beam footprint's center and that of the receiver aperture, r, is approximated as

$$\eta_{\text{p}} \approx A_0 \exp\left(-\frac{2r^2}{w_{L_{\text{eq}}}^2}\right), \tag{6}$$

where $A_0 = [\text{erf}(\nu)]^2$ denotes the maximum fraction of collected power, $\nu = \sqrt{\pi}D_{\text{r}}/\left(2\sqrt{2}w_{L,\text{eff}}\right)$, D_{r} is the receiver aperture's diameter, and $w_{L_{\text{eq}}}^2 = w_{L,\text{eff}}^2 \frac{\sqrt{\pi}\text{erf}(\nu)}{2\nu\exp(-\nu^2)}$ is the equivalent beam width. Additionally, $w_{L,\text{eff}} = w_L\sqrt{1+G_{\text{d}}}$ is the effective beam width, $w_L = w_0\sqrt{\left(1 - \frac{L}{F_0}\right)^2 + \left(\frac{L\lambda}{\pi w_0^2}\right)^2}$, $L = (h_{\text{HAP}} - h_{\text{UAV}})\sec(\zeta)$ is the slant distance, h_{HAP} is the HAP's altitude, G_{d} can be found in [], F_0 is the radius of curvature, $w_0 = \frac{2\lambda}{\pi\theta}$, and θ is the divergence full-angle.

In this work, we consider two factors that contribute to the misalignment model of r, which are (1) the operation and hovering of the UAV within the beam footprint, and (2) the satellite's vibration. For the first factor, the UAV is not necessarily at the center of the beam footprint. Thus, the radial displacement vector can be expressed as $\mathbf{r}_1 = (X_1, Y_1)$ where $X_1 \sim \mathcal{N}\left(\rho_{\text{x}}, \sigma_{\text{h}}^2\right)$, and $Y_1 \sim \mathcal{N}\left(\rho_{\text{y}}, \sigma_{\text{h}}^2\right)$. Regarding the second factor, the radial displacement vector in this case can be given as $\mathbf{r}_2 = (X_2, Y_2)$, where $X_2 \sim \mathcal{N}\left(0, \sigma_{\text{jt}}^2\right)$, $Y_2 \sim \mathcal{N}\left(0, \sigma_{\text{jt}}^2\right)$, $\sigma_{\text{jt}} \simeq L\theta_{\text{jt}}$, and θ_{jt} denotes the jitter angle. As a result, $r = \|\mathbf{r}_1 + \mathbf{r}_2\|_2$ follows the Rician distribution, whose PDF is given as [3]

$$f_r(r) = \frac{r}{\sigma_{\text{s}}^2} \exp\left(-\frac{r^2+\rho^2}{2\sigma_{\text{s}}^2}\right) I_0\left(\frac{\rho r}{\sigma_{\text{s}}^2}\right), \tag{7}$$

where $\rho = \sqrt{\rho_{\text{x}}^2 + \rho_{\text{y}}^2}$, $\sigma_{\text{s}}^2 = \sigma_{\text{h}}^2 + \sigma_{\text{jt}}^2$, and $I_0(\cdot)$ is the modified Bessel function of the first kind with order zero. Combining (6) and (7), we obtain the PDF of η_{p} as [3]

$$f_{\eta_{\text{p}}}(\eta_{\text{p}}) = \frac{\chi^2}{A_0^{\chi^2}} \exp\left(\frac{-\rho^2}{2\sigma_{\text{s}}^2}\right) \eta_{\text{p}}^{\chi^2-1} I_0\left(\frac{\rho}{\sigma_{\text{s}}^2}\sqrt{\frac{-w_{L_{\text{eq}}}^2}{2}\ln\frac{\eta_{\text{p}}}{A_0}}\right), \tag{8}$$

where $0 \le \eta_{\text{p}} \le A_0$, and $\chi = w_{L_{\text{eq}}}/2\sigma_{\text{s}}$.

Regarding the received signal's AoA fluctuation, it arises from the UAV's angular instability and can cause signal loss

due to the receiver's limited field of view (FOV). Consequently, the PDF of η_{AoA} is given as [3]

$$f_{\eta_{\text{AoA}}}(\eta_{\text{AoA}}) = P_0\delta(\eta_{\text{AoA}}) + P_1\delta(\eta_{\text{AoA}} - 1), \quad (9)$$

where $P_0 = \exp\left(-\frac{\theta_{\text{FoV}}^2}{2\sigma_{\text{o}}^2}\right)$, $P_1 = 1 - P_0$, θ_{FoV} is the detector's FOV half-angle, and σ_{o} denotes the standard deviation (SD) of angular fluctuation.

Utilizing (4), (5), (8), and (9) and applying some mathmatical transformations, we derive the PDF of η as [3]

$$f_\eta(\eta) = P_0\delta(\eta) + P_1\frac{\chi^2 C_1}{2(A_0\eta_{\text{atm}})^{\chi^2}}\eta^{\chi^2-1}\text{erfc}\left[C_2(\eta)\right], \quad (10)$$

where $C_1 = \exp\left[\frac{\rho^2}{\sigma_{\text{s}}^2} + 0.5\sigma_{\text{R}}^2\chi^2\left(1+\chi^2\right)\right]$, $C_2(\eta) = C_3\left[\ln\left(\frac{\eta}{A_0\eta_{\text{atm}}}\right) + \mu\right]$, $C_3 = \left(\frac{32\rho^2\sigma_{\text{s}}^2}{w_{L_{\text{eq}}}^4} + 2\sigma_{\text{R}}^2\right)^{-\frac{1}{2}}$ and $\mu = \frac{6\rho^2}{w_{L_{\text{eq}}}^2} + 0.5\sigma_{\text{R}}^2\left(1+2\chi^2\right)$.

III. Secret-key Rate Analysis & Optimization

A. Secret-key Rate Analysis

The secret-key rate can be computed as [4]

$$\text{SKR} = \beta I_{\text{AB}} - \chi_{\text{BE}}, \quad (11)$$

where β is the key reconciliation efficiency, I_{AB} is the mutual information between Alice and Bob, and χ_{BE} is the Holevo bound representing the maximum information acquired by Eve. The mutual information between Alice and Bob for M-ary modulation is generally given as [5]

$$I_{\text{AB}} = \sum_{\mathbf{k}\in\mathcal{K}} p_{\mathbf{k}} \int p(y|\alpha_{\mathbf{k}}) \log_2\left[\frac{p(y|\alpha_{\mathbf{k}})}{\sum_{\mathbf{k}'\in\mathcal{K}} p_{\mathbf{k}'} p(y|\alpha_{\mathbf{k}'})}\right] \mathrm{d}y, \quad (12)$$

where $p(y|\alpha_{\mathbf{k}}) = \frac{1}{\sqrt{2\pi\sigma_\xi^2}}\exp\left[-\frac{\left|y-2\sqrt{\eta_{\text{f}}}\alpha_{\mathbf{k}}\right|^2}{2\sigma_\xi^2}\right]$ and $\sigma_\xi^2 = 1 + \eta_{\text{f}}\xi_{\text{f}}$. Moreover, η_{f} and ξ_{f} are the effective transmittance and effective excess noise, respectively, whose calculations are given at the end of this Section.

As for the Holevo bound, it can be computed based on the symplectic eigenvalues $\{\nu_i\}_{i=1}^3$ of the covariance matrix as [2]

$$\chi_{\text{BE}} = g\left(\frac{\nu_1 - 1}{2}\right) + g\left(\frac{\nu_2 - 1}{2}\right) - g\left(\frac{\nu_3 - 1}{2}\right), \quad (13)$$

where $g(x) = (x+1)\log_2(x+1) - x\log_2(x)$, $\nu_{1,2} = \sqrt{\frac{\Delta \pm \sqrt{\Delta^2 - 4D}}{2}}$, and $\nu_3 = \sqrt{V\left(V - \frac{Z^2}{W}\right)}$. Moreover, $\Delta = V^2 + W^2 - 2Z^2$, $D = \left(VW - Z^2\right)^2$, $V = V_{\text{mod}} + 1$, $W = 1 + \eta_{\text{f}}V_{\text{mod}} + \eta_{\text{f}}\xi_{\text{f}}$, and Z is given as

$$Z(\eta_{\text{f}}, \xi_{\text{f}}) = 2\sqrt{\eta_{\text{f}}}\text{tr}\left(\tau^{1/2} a \tau^{1/2} a^\dagger\right) - \sqrt{2\eta_{\text{f}}\xi_{\text{f}} w}. \quad (14)$$

Therein, τ denotes the density matrix and can be given as [2]

$$\tau = \sum_{\mathbf{k}\in\mathcal{K}} p_{\mathbf{k}} |\alpha_{\mathbf{k}}\rangle\langle\alpha_{\mathbf{k}}|, \quad (15)$$

where a coherent state can be approximated by a finite superposition of Fock states as $|\alpha\rangle \approx e^{\frac{-|\alpha|^2}{2}} \sum_{n=0}^{N_{\max}} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$ [2]. Moreover, a and $a^\dagger$ are the annihilation and creation operators, respectively, and w is a quantity measuring how far the finite constellation is from Gaussian modulation, and is given by [2]

$$w = \sum_k p_k \left(\langle\alpha_k|a_\tau^\dagger a_\tau|\alpha_k\rangle - |\langle\alpha_k|a_\tau|\alpha_k\rangle|^2\right), \quad (16)$$

where $a_\tau = \tau^{1/2} a \tau^{-1/2}$.

Regarding the effective transmittance and effective excess noise, these parameters are expressed as [6]

$$\eta_{\text{f}} = \langle\sqrt{\eta}\rangle^2, \quad \eta_{\text{f}}\xi_{\text{f}} = \text{Var}(\sqrt{\eta})V_{\text{mod}} + \langle\eta\xi\rangle, \quad (17)$$

where $\text{Var}(\sqrt{\eta}) = \langle\eta\rangle - \langle\sqrt{\eta}\rangle^2$, $\langle\eta\rangle = \int_0^1 \eta f_\eta(\eta)\,\mathrm{d}\eta$, $\langle\sqrt{\eta}\rangle = \int_0^1 \sqrt{\eta} f_\eta(\eta)\,\mathrm{d}\eta$, and $\langle\eta\xi\rangle = \int_0^1 \eta\xi f_\eta(\eta)\,\mathrm{d}\eta$. Furthermore, $\xi = \xi_{\text{ch}} + \frac{\xi_{\text{det}}}{\eta}$ is the total excess noise, ξ_{ch} is the channel excess noise, and ξ_{det} represents the detector excess noise [4].

Lemma 1. *The expressions of $\langle\eta\rangle$, $\langle\sqrt{\eta}\rangle$, and $\langle\eta\xi\rangle$ are respectively written as*

$$\langle\eta\rangle = P_1 A_0 \eta_{\text{atm}} \frac{\chi^2 C_1}{\chi^2 + 1} \mathcal{Q}\left(\chi^2 + 1\right), \quad (18)$$

$$\langle\sqrt{\eta}\rangle = P_1\sqrt{A_0\eta_{\text{atm}}} \frac{\chi^2 C_1}{\chi^2 + 0.5} \mathcal{Q}\left(\chi^2 + 0.5\right), \quad (19)$$

$$\langle\eta\xi\rangle = \xi_{\text{ch}}\langle\eta\rangle + \xi_{\text{det}}\left[P_0 + P_1 C_1 \mathcal{Q}\left(\chi^2\right)\right], \quad (20)$$

where $\mathcal{Q}(a) = 0.5\left[\mathcal{Q}_1(a) + \mathcal{Q}_2(a)\right]$, $\mathcal{Q}_1(a) = \exp(ax_1 - a\mu)\,\text{erfc}(C_3 x_1)$, $x_1 = \ln\left(\frac{1}{A_0\eta_{\text{atm}}}\right) + \mu$, *and* $\mathcal{Q}_2(a) = \exp\left(\frac{a^2}{4C_3^2} - a\mu\right)\text{erfc}\left(\frac{a}{2C_3} - C_3 x_1\right)$.

Proof. Let $B = A_0\eta_{\text{atm}}$. By applying some mathematical manipulations, we can derive $\langle\eta\rangle$ as

$$\langle\eta\rangle = P_1 \frac{\chi^2 C_1}{2B^{\chi^2}} \int_0^1 \eta^{\chi^2} \text{erfc}\left[C_3\left(\ln\frac{\eta}{B} + \mu\right)\right] \mathrm{d}\eta. \quad (21)$$

By applying the change of variables $x = \ln\frac{\eta}{B} + \mu$, integrating by parts, and utilizing the definition of the complementary error function [], we obtain the expression for $\langle\eta\rangle$ as in (18). The expressions of $\langle\sqrt{\eta}\rangle$ and $\langle\eta\xi\rangle$ can be derived by applying the same approach. This completes the proof. ■

B. Secret-key Rate Optimization

TBD

IV. Numerical Results & Discussions

Unless otherwise noted, default parameters used in this paper are as follows. *At the HAP (Alice):* HAP altitude $h_{\text{HAP}} = 20$ km, zenith angle $\zeta = 30°$, divergence full-angle $\theta = 233$ μrad, $F_0 = \infty$, optical wavelength $\lambda = 1550$ nm, jitter angle $\theta_{\text{jt}} = 15.61$ μrad. *At the UAV (Bob):* UAV altitude $h_{\text{UAV}} = 1$ km, receiver aperture diameter $D_{\text{r}} = 8$ cm, boresight offsets $\rho_{\text{x}} = \rho_{\text{y}} = 0$ m, SD of UAV position $\sigma_{\text{h}} = 1$ m, SD of UAV's angular fluctuation $\sigma_{\text{o}} = 1.5$ mrad, the detector's FOV half-angle $\theta_{\text{FoV}} = 6.25$ mrad. *Regarding the quantum channel:* atmospheric altitude $h_{\text{atm}} = 20$ km, atmospheric transmittance $\tau_{\text{zen}} = 0.92$, ground turbulence level $C_{\text{n}}^2(0) = 10^{-14}$ m$^{-2/3}$, rms wind speed $v_{\text{wind}} = 21$ m/s.

key reconciliation efficiency $\beta = 0.95$

REFERENCES

[1] M. N. Notarnicola, S. Olivares, E. Forestieri, E. Parente, L. Potì, and M. Secondini, "Probabilistic amplitude shaping for continuous-variable quantum key distribution with discrete modulation over a wiretap channel," *IEEE Trans. Commun.*, vol. 72, no. 1, pp. 375–386, Jan. 2024.

[2] A. Denys, P. Brown, and A. Leverrier, "Explicit asymptotic secret key rate of continuous-variable quantum key distribution with an arbitrary modulation," *Quantum*, vol. 5, p. 540, Sep. 2021.

[3] H. D. Le *et al.*, "Throughput analysis for TCP over the FSO-based satellite-assisted internet of vehicles," *IEEE Trans. Veh. Technol.*, vol. 71, no. 2, pp. 1875–1890, Feb. 2022.

[4] F. Laudenbach *et al.*, "Continuous-variable quantum key distribution with gaussian modulation—the theory of practical implementations," *Adv. Quantum Technol.*, vol. 1, no. 1, p. 1800011, Jun. 2018.

[5] T. K. Moon, *Error correction coding: Mathematical Methods and Algorithms*. John Wiley & Sons, 2020.

[6] N. Hosseinidehaj, N. Walk, and T. C. Ralph, "Composable finite-size effects in free-space continuous-variable quantum-key-distribution systems," *Phys. Rev. A*, vol. 103, no. 1, p. 012605, Jan. 2021.